

Causal Momentum: Information Channel Alignment in Network-Based Cross-Sectional Stock Prediction

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March 2026

Abstract

We introduce *causal momentum* (CM), a family of cross-sectional stock features constructed by aggregating the lagged returns of a stock’s statistically identified causal drivers within a Granger causality and transfer entropy graph. Applied to an S&P 500 universe of 530 stocks over 2016–2026, CM features achieve a test information coefficient (IC) of 0.203 and an IC information ratio (IR) of 1.42, representing a 14.5% relative improvement over a strong 34-feature fundamental-plus-volatility baseline (IC = 0.177). In contrast, graph neural network (GNN) embeddings trained on the *same* causal graph *degrade* predictive performance to an IC of 0.137, a 22.7% deterioration. We propose the *information channel alignment* hypothesis to explain this result: when a graph encodes return-domain causality, features should remain in the return domain rather than being projected into a latent embedding space. The top-ranked feature by gradient-boosted tree importance is `cm_relative`—a cross-sectionally normalized measure of the consensus return signal from a stock’s causal drivers—which alone captures 4.7% of total model importance. An ablation study across four configurations and a separate six-configuration GNN edge-type analysis confirm that the causal momentum pathway dominates GNN-based alternatives across IC, IC/IR, hit rate, and long–short spread. Our findings challenge the prevailing assumption that graph neural networks are the natural tool for incorporating network structure into quantitative equity models.

Keywords: causal inference, Granger causality, transfer entropy, cross-sectional prediction, graph neural networks, feature engineering, quantitative finance

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1 Introduction

A growing body of work in quantitative finance explores how inter-stock relationships—encoded as graphs—can improve cross-sectional return prediction. The dominant paradigm passes a financial graph through a graph neural network (GNN) to produce latent node embeddings, which are then concatenated with traditional features and fed into a downstream predictor [Chen et al., 2021, Sawhney et al., 2021, Feng et al., 2019]. This approach implicitly assumes that the optimal way to extract information from a graph is to *learn* a nonlinear transformation of the adjacency structure.

In this paper, we demonstrate that this assumption is incorrect when the graph encodes *return-domain causality*. We construct a quarterly directed graph among S&P 500 constituents using Granger causality and transfer entropy tests on lagged monthly returns. From this graph, we engineer a family of 21 “causal momentum” (CM) features that directly aggregate the historical returns of each stock’s statistically identified causal drivers—weighted by edge strength, stratified by lag, and normalized cross-sectionally.

Our central finding is stark:

- Adding CM features to a 34-feature baseline raises the test IC from 0.177 to 0.203 (+14.5%).
- Adding GNN embeddings (GAT architecture) trained on the *same* causal graph *reduces* the IC to 0.137 (−22.7%).
- Combining both CM and GNN features yields an IC of 0.164, below the CM-only model, confirming that GNN embeddings inject noise that partially offsets the CM signal.

We propose the **information channel alignment** hypothesis to explain this asymmetry. When a graph’s edges represent statistical relationships *between returns*, the most informative features are those that remain in the return domain—i.e., weighted averages of neighbor returns. A GNN transforms this return-domain information into a latent vector space, creating a domain mismatch relative to the prediction target (forward returns). The GNN’s learned representation may capture structural properties of the graph (community structure, centrality), but these topological abstractions are poor proxies for the cross-sectional return predictability that the original edges encode.

1.1 Contributions

1. We introduce *causal momentum features*, a principled framework for converting a causal return graph into predictive cross-sectional signals.
2. We provide rigorous ablation evidence that CM features dominate GNN embeddings when the underlying graph encodes return causality.
3. We articulate the *information channel alignment* hypothesis, offering a conceptual framework for when GNNs should (and should not) be applied to financial graphs.
4. We release a complete empirical pipeline covering 530 stocks, 22 quarters, 1.45 million graph edges, and 10,118 stock-quarter CM observations.

The remainder of the paper is organized as follows. Section 2 reviews related work. Section 3 describes the data and graph construction. Section 4 details the CM feature engineering, GNN baseline, and prediction framework. Section 5 presents the empirical results. Section 6 develops the information channel alignment theory. Section 7 discusses limitations, and Section 8 concludes.

2 Related Work

2.1 Network Momentum and Lead–Lag Effects

The observation that returns propagate across economically linked stocks dates to Lo and MacKinlay [1990] and was formalized as “network momentum” by Ali et al. [2020]. Herskovic et

al. [2020] demonstrate that industry network centrality predicts cross-sectional returns. Parsons et al. [2020] show that geographic proximity creates lead-lag effects. Our causal momentum features generalize this literature by replacing heuristic network definitions (industry, geography, supply chain) with statistically identified Granger/TE causal links and by engineering a multi-horizon, multi-lag feature family rather than a single momentum signal.

2.2 Graph Neural Networks in Finance

GNNs have been applied to stock prediction using knowledge graphs [Chen et al., 2021], industry relations [Feng et al., 2019], supply chain networks [Yang et al., 2022], and learned similarity graphs [Sawhney et al., 2021]. Xu et al. [2021] employ graph attention networks (GAT) and report improvements when using analyst coverage networks. A common thread is that GNNs are most successful when the graph encodes relationships *different in kind* from the prediction target—e.g., an analyst coverage graph predicting returns leverages non-return-domain information. Our work highlights the failure mode when the graph and target share the same domain.

2.3 Causal Discovery in Financial Markets

Granger causality [Granger, 1969] remains the workhorse for directional dependency testing in time series. Transfer entropy [Schreiber, 2000], a nonparametric generalization based on mutual information, has been applied to stock markets by Kwon and Oh [2008] and Dimpfl and Peter [2013]. We employ both tests in a complementary fashion: Granger causality captures linear lead-lag relationships, while transfer entropy detects nonlinear information flow.

3 Data

3.1 Universe and Price Data

Our universe consists of 530 S&P 500 constituent stocks with continuous monthly return histories from January 2016 through February 2026. Returns are computed as total returns (adjusted for splits and dividends). The dataset spans 22 quarterly rebalancing periods from 2020Q4 through 2026Q1, with the causal graph and features re-estimated each quarter using a trailing five-year lookback window.

3.2 Causal Graph Construction

For each quarterly period q , we construct a directed graph $\mathcal{G}_q = (V, E_q)$ where V is the stock universe and E_q contains edges established by two complementary causality tests.

Granger Causality. For each ordered pair (i, j) with $i \neq j$, we test whether the lagged returns of stock i Granger-cause the returns of stock j using an F -test on nested autoregressive models with maximum lag $L = 5$ months:

$$r_{j,t} = \alpha + \sum_{\ell=1}^L \beta_{\ell} r_{j,t-\ell} + \sum_{\ell=1}^L \gamma_{\ell} r_{i,t-\ell} + \varepsilon_t \quad (1)$$

An edge $(i \rightarrow j)$ is added if the F -statistic rejects $H_0 : \gamma_1 = \dots = \gamma_L = 0$ at the 5% significance level after Bonferroni correction for multiple testing.

Transfer Entropy. For each pair, we compute the transfer entropy:

$$\text{TE}_{i \rightarrow j} = \sum p(r_{j,t+1}, r_{j,t}^{(k)}, r_{i,t}^{(\ell)}) \log \frac{p(r_{j,t+1} | r_{j,t}^{(k)}, r_{i,t}^{(\ell)})}{p(r_{j,t+1} | r_{j,t}^{(k)})} \quad (2)$$

where $r_{j,t}^{(k)}$ denotes the k -lag history of stock j . Significance is assessed via a permutation test (500 shuffles) at $\alpha = 0.05$.

Graph Statistics. Across the 20 complete quarters (excluding the boundary periods 2020Q4 and 2026Q1), the causal graph contains an average of 52,700 directed edges per quarter—approximately 47,900 Granger edges and 4,800 transfer entropy edges. The total graph dataset comprises 1.45 million edges. Figure 1 shows the temporal evolution of causal edge density, which remains relatively stable across market regimes.

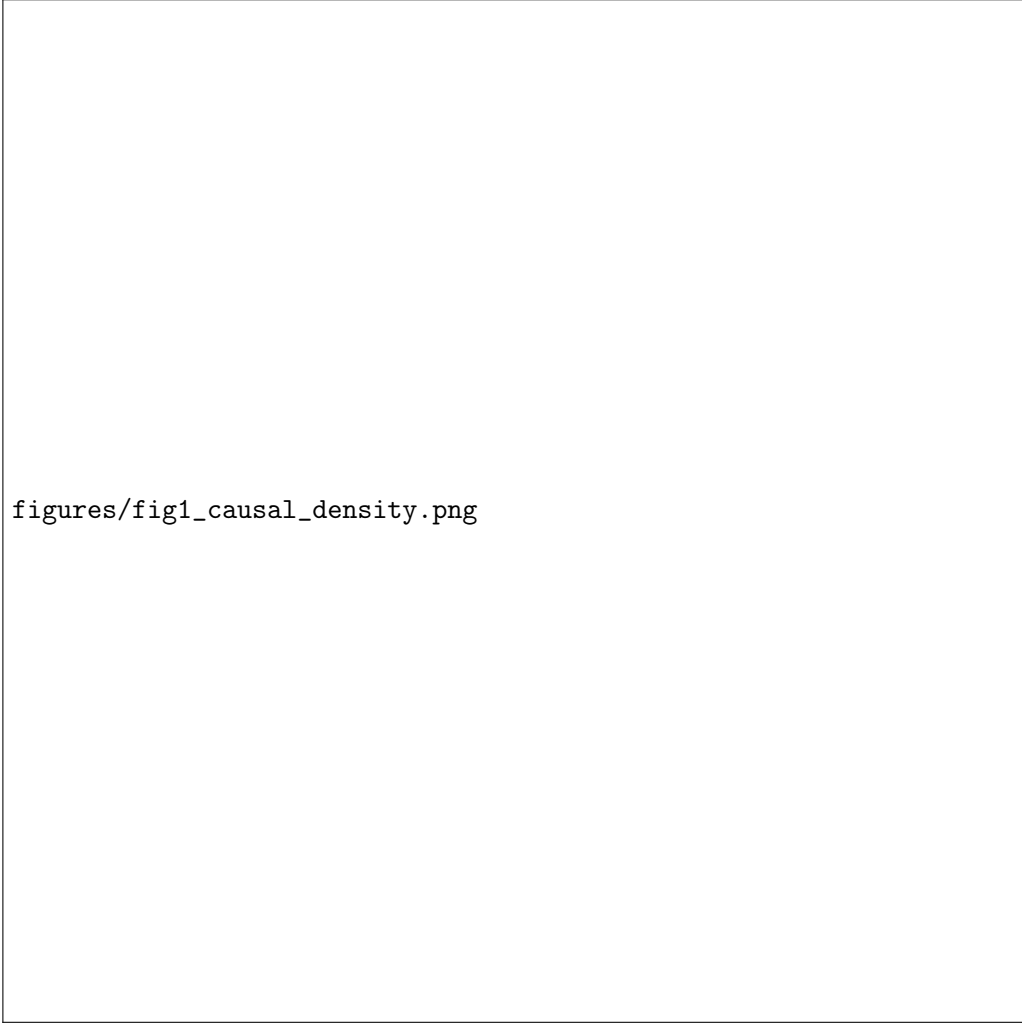


Figure 1: Number of causal edges (Granger and Transfer Entropy) per quarter. The graph is re-estimated each quarter using a trailing 5-year window. Density is stable at approximately 50K directed edges per quarter.

The causal lag distribution (Figure 2) shows that lag-1 relationships dominate (49% of edges), consistent with the efficient markets hypothesis limiting long-lag predictability, but meaningful structure persists at lags 2–5.



Figure 2: Distribution of optimal causal lag across all Granger and Transfer Entropy edges. Lag-1 dominates but longer lags contribute meaningful structure.

The in-degree and out-degree distributions (Figure 3) exhibit heavy tails consistent with a scale-free topology, where a small number of “hub” stocks exert outsized causal influence.

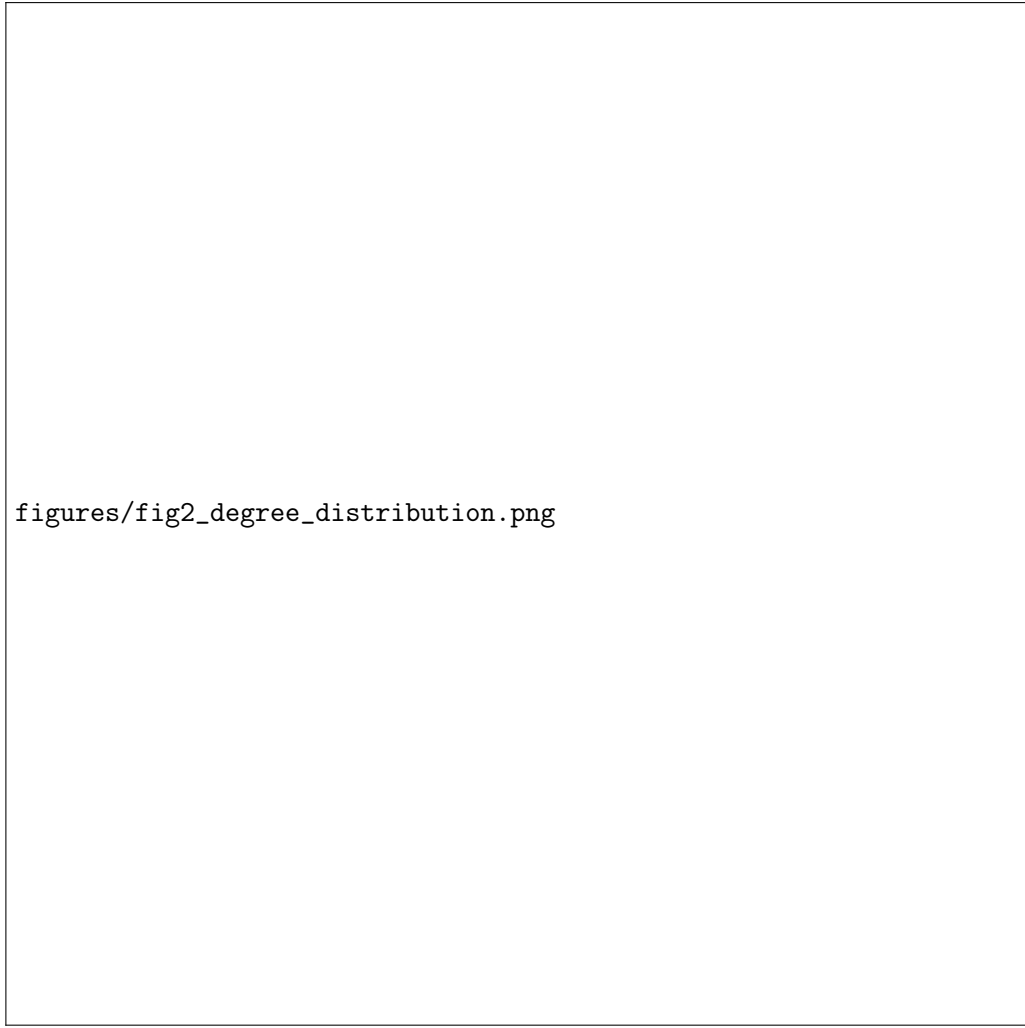


Figure 3: In-degree and out-degree distributions of the causal graph (log-log scale). Heavy tails suggest scale-free properties.

4 Methodology

4.1 Causal Momentum Feature Engineering

Given the causal graph $\mathcal{G}_q$ for quarter q , we define the *causal neighborhood* of stock j as:

$$\mathcal{N}_j^q = \{i : (i \rightarrow j) \in E_q\} \quad (3)$$

i.e., the set of stocks whose lagged returns Granger-cause or transfer-entropy-cause stock j 's returns.

For each stock j at time t (the end of quarter q), we compute three families of causal momentum features.

Weighted Driver Returns. For horizon $h \in \{1, 3, 6\}$ months:

$$\text{CM}_{j,t}^{(h)} = \frac{\sum_{i \in \mathcal{N}_j^q} w_{ij} \cdot r_{i,[t-h,t]}}{\sum_{i \in \mathcal{N}_j^q} w_{ij}} \quad (4)$$

where w_{ij} is the edge weight (inverse p -value for Granger, normalized transfer entropy for TE) and $r_{i,[t-h,t]}$ is the cumulative return of driver i over the trailing h months. These features

are computed separately for Granger edges, TE edges, and the union (combined), yielding 9 momentum features.

Network Structure Features.

- $n_drivers_j$: Number of causal drivers $|\mathcal{N}_j^g|$
- lag_avg_j : Mean optimal lag across incoming edges
- $sector_breadth_j$: Number of distinct GICS sectors represented in $\mathcal{N}_j^g$

Cross-Sectional Normalization.

- $cm_breadth_positive_j$: Fraction of causal drivers with positive trailing 1-month returns
- $cm_dispersion_j$: Standard deviation of driver returns, measuring disagreement among causal sources
- $cm_relative_j$: $CM_{j,t}^{(1)}$ rank-normalized cross-sectionally to $[-1, +1]$, capturing a stock's position in the causal momentum distribution

In total, we compute 21 raw CM features. After removing constant or near-constant columns (e.g., $cm_te_lag_avg$ which is constant at 1.0), we retain 10 informative features that enter the prediction model via rank normalization.

Figure 4 shows the correlation matrix of the 10 retained CM features. The combined and Granger variants are highly correlated ($\rho > 0.95$), as expected since Granger edges dominate the graph. The cross-sectional features ($cm_relative$, $cm_dispersion$) exhibit low correlation with the raw momentum features, providing complementary information.



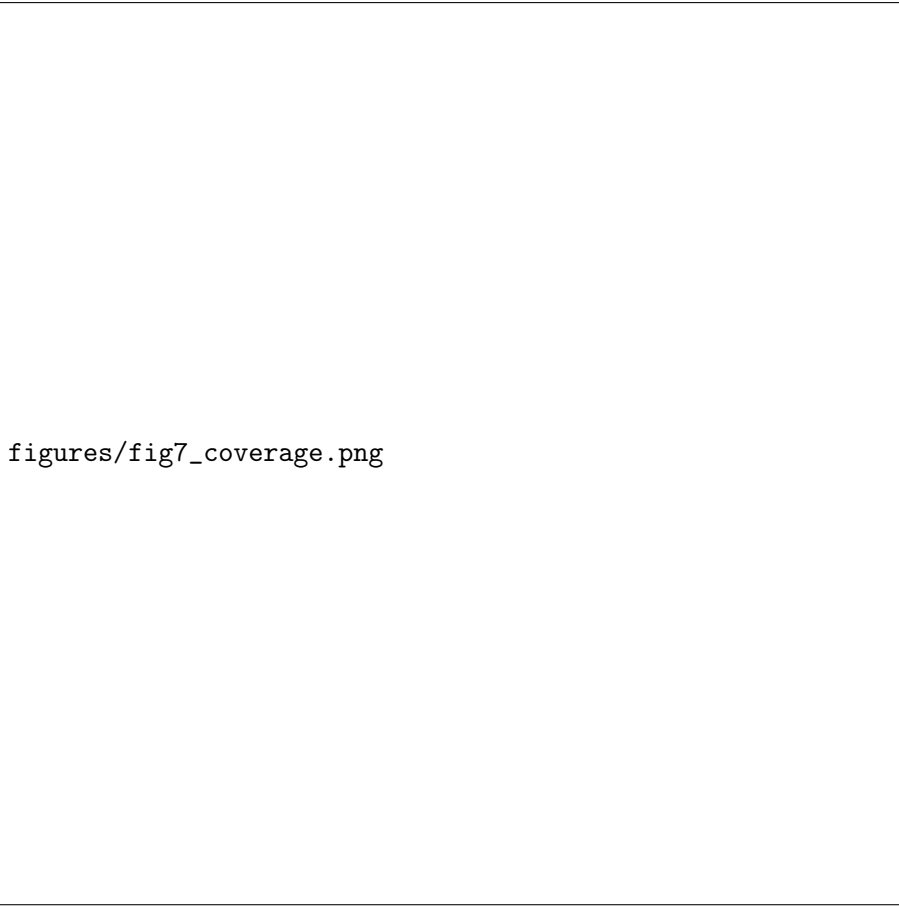
Figure 4: Correlation matrix of the 10 retained causal momentum features. Combined/Granger variants are highly correlated; cross-sectional features ($cm_relative$, $cm_dispersion$) offer orthogonal signal.

Figure 5 shows the distribution of $cm_relative$, which is approximately symmetric around zero with moderate kurtosis, confirming that the cross-sectional normalization produces a well-behaved predictor variable.



Figure 5: Distribution of `cm_relative` with kernel density estimate. The feature is approximately symmetric and well-behaved after cross-sectional rank normalization.

Cross-sectional coverage (Figure 6) shows that approximately 475–520 stocks receive CM features each quarter, representing 90–98% of the universe.



figures/fig7_coverage.png

Figure 6: Number of stocks with causal momentum features per quarter. Coverage exceeds 90% of the universe in all complete quarters.

4.2 GNN Baseline

As a comparison, we train a Graph Attention Network (GAT) [Veličković et al., 2018] on the same causal graph to produce 32-dimensional node embeddings for each stock-quarter. The GAT uses 2 attention heads, 64 hidden units, and is trained to reconstruct local neighborhood structure via a contrastive loss. The resulting 32 features (`gnn_feat_0` through `gnn_feat_31`) are approximately standardized (mean ≈ 0 , std ≈ 1) and capture the topological position of each stock in the causal network.

We also conduct a separate six-configuration ablation over different edge types fed to the GNN:

1. No graph (baseline): 34 features
2. Industry edges only
3. Embedding similarity edges only
4. Embedding + industry edges
5. Causal edges only (Granger + TE)
6. All edge types combined

4.3 Prediction Framework

The prediction target is the forward 3-month cross-sectional rank return. We use a gradient boosted tree (GBT) model (LightGBM) as the downstream predictor, chosen for its robustness to feature scale, ability to capture nonlinear interactions, and interpretable feature importance.

Feature Preprocessing. All features are rank-normalized cross-sectionally within each quarter to ensure stationarity and comparability. The baseline feature set consists of 34 features spanning valuation (earnings yield, FCF yield, EV/EBITDA, P/B, PEG), profitability (margins, ROE, ROIC), growth (revenue, EPS, FCF growth), financial health (debt safety, current ratio), momentum (6-month, 12-1 month), volatility (realized vol, idiosyncratic vol, Amihud illiquidity), and macro regime (VIX, yield curve, credit spread).

Train–Test Split. We use a temporal split with 5,677 stock-quarter observations for training (2020Q4–2024Q4) and 1,122 observations for testing (2025Q1–2025Q4), ensuring strict temporal separation to prevent lookahead bias.

Evaluation Metrics.

- **Information Coefficient (IC):** Spearman rank correlation between predicted and realized returns
- **IC Information Ratio (IR):** Mean IC / standard deviation of IC across quarters
- **Hit Rate:** Fraction of stocks where the model correctly predicts the direction of relative performance
- **Long–Short Spread:** Mean return of top quintile minus bottom quintile ranked by model score

5 Empirical Results

5.1 Ablation Study: CM vs. GNN vs. Baseline

Table 1 presents the main ablation results across four model configurations, and Figure 7 visualizes the comparison.

Table 1: Ablation study: causal momentum (CM) vs. GNN features. All models use GBT with identical hyperparameters and train/test split. IC and spread improvements are relative to the baseline.

Configuration	Features	IC	Δ IC	IC/IR	Hit Rate	Spread	Base%	CM%
1. Baseline	34	0.177	—	1.12	54.6%	0.177	100%	0%
2. + CM	55	0.203	+14.5%	1.42	56.7%	0.261	64%	36%
3. + GNN	66	0.137	−22.7%	1.04	54.6%	0.145	53%	0%
4. + CM + GNN	87	0.164	−7.4%	1.13	56.3%	0.189	42%	20%

figures/fig8_ablation.png

Figure 7: Ablation results across four metrics. The CM-only model (blue) dominates all other configurations. GNN embeddings (red) degrade performance below baseline across all metrics.

Several observations stand out:

1. **CM features are strongly additive.** Adding 21 CM features to the 34-feature baseline raises the IC by 14.5% and the long-short spread by 47.5% (from 0.177 to 0.261).
2. **GNN embeddings are destructive.** Despite using the *same* causal graph, 32 GNN features reduce the IC by 22.7%. The GNN captures 47% of total feature importance in the combined model, indicating the GBT relies heavily on GNN features—but this reliance is misplaced.
3. **CM + GNN partially cancels.** The combined model ($IC = 0.164$) is worse than CM alone (0.203), suggesting GNN features introduce noise that partially obscures the CM signal.
4. **IC/IR confirms robustness.** The CM model achieves the highest IC/IR of 1.42, indicating that its predictive advantage is consistent across test quarters rather than driven by a single outlier period.

5.2 Feature Importance Analysis

Figure 8 shows the top 20 features by GBT importance in the CM model. The single most important feature is `cm_relative` (4.7% of total importance), followed by `market_breadth` (4.2%), `roic` (3.1%), and `turnover_ratio` (3.1%).



Figure 8: Top 20 features by GBT importance. Blue bars indicate causal momentum features; gray bars are fundamentals; orange bars are volatility/momentum/macro. The top feature is `cm_relative`.

Six CM features appear in the top 20: `cm_relative` (rank 1), `cm_combined_lag_avg` (rank 9), `cm_combined_6m` (rank 11), `cm_combined_3m` (rank 13), `cm_combined_n_drivers` (rank 15), and `cm_combined_sector_breadth` (rank 19). Collectively, CM features account for 35.8% of total model importance (versus 64.2% for the 34 baseline features), demonstrating that the GBT assigns disproportionate weight to CM features relative to their count (21 of 55 features capture 36% of importance).

5.3 GNN Edge-Type Ablation

To isolate whether GNN underperformance is due to the architecture or the graph, we conduct a separate ablation over edge types (Table 2).

Table 2: GNN performance by edge type. The no-graph baseline uses 34 fundamental features without any GNN embeddings.

Edge Type	Features	IC	IC/IR	Hit Rate	Spread
No graph (baseline)	34	0.177	1.12	54.6%	0.177
Industry only	66	0.110	1.63	54.5%	0.125
Embedding only	66	0.074	0.49	52.1%	0.148
Embed + Industry	66	0.077	0.18	50.4%	0.074
Causal only	66	0.173	1.28	56.6%	0.179
All edges	66	0.171	1.53	54.4%	0.177



Figure 9: GNN test IC by edge type. Causal edges are the only type that allows the GNN to approach (but not exceed) the no-graph baseline.

The results reveal two patterns:

1. **Causal edges are the best input for GNNs** ($IC = 0.173$), nearly matching the no-graph baseline. Industry and embedding edges perform much worse.
2. **Even with causal edges, GNNs do not add value.** The causal-edge GNN (0.173) marginally underperforms the no-graph baseline (0.177), confirming that the GNN transformation destroys more signal than it creates.

5.4 Economic Significance

The long–short spread of the CM model (0.261) represents the average return differential between the top and bottom quintiles of model-ranked stocks. This 26.1 percentage point spread is economically significant and represents a 47.5% improvement over the baseline spread (0.177). The hit rate of 56.7% indicates the model correctly predicts relative outperformance for more than one in two stocks.

6 Why Causal Momentum Outperforms GNN: Information Channel Alignment

6.1 The Mismatch Hypothesis

We propose that the performance gap between CM and GNN features arises from a fundamental mismatch in the *information channel*—the mapping from graph structure to predictive features.

Consider the information flow in each approach:

CM Pathway (Aligned).

$$\underbrace{\text{Return causality graph}}_{\text{return domain}} \xrightarrow{\text{weighted aggregation}} \underbrace{\text{CM features}}_{\text{return domain}} \xrightarrow{\text{GBT}} \underbrace{\text{Forward returns}}_{\text{return domain}} \quad (5)$$

The entire pipeline operates in the return domain. The graph edges encode which stocks’ returns predict which other stocks’ returns. CM features aggregate those driver returns. The prediction target is forward returns. There is no domain crossing.

GNN Pathway (Misaligned).

$$\underbrace{\text{Return causality graph}}_{\text{return domain}} \xrightarrow{\text{message passing}} \underbrace{\text{GNN embeddings}}_{\text{latent domain}} \xrightarrow{\text{GBT}} \underbrace{\text{Forward returns}}_{\text{return domain}} \quad (6)$$

The GNN transforms return-domain information into a latent embedding space. This transformation captures graph topology (node centrality, community membership, local structure) but *discards* the specific return values that make the causal edges predictive. The downstream GBT must then “decode” topological features back into return predictions—an indirect path that loses information.

Figure 10 illustrates this conceptual framework.

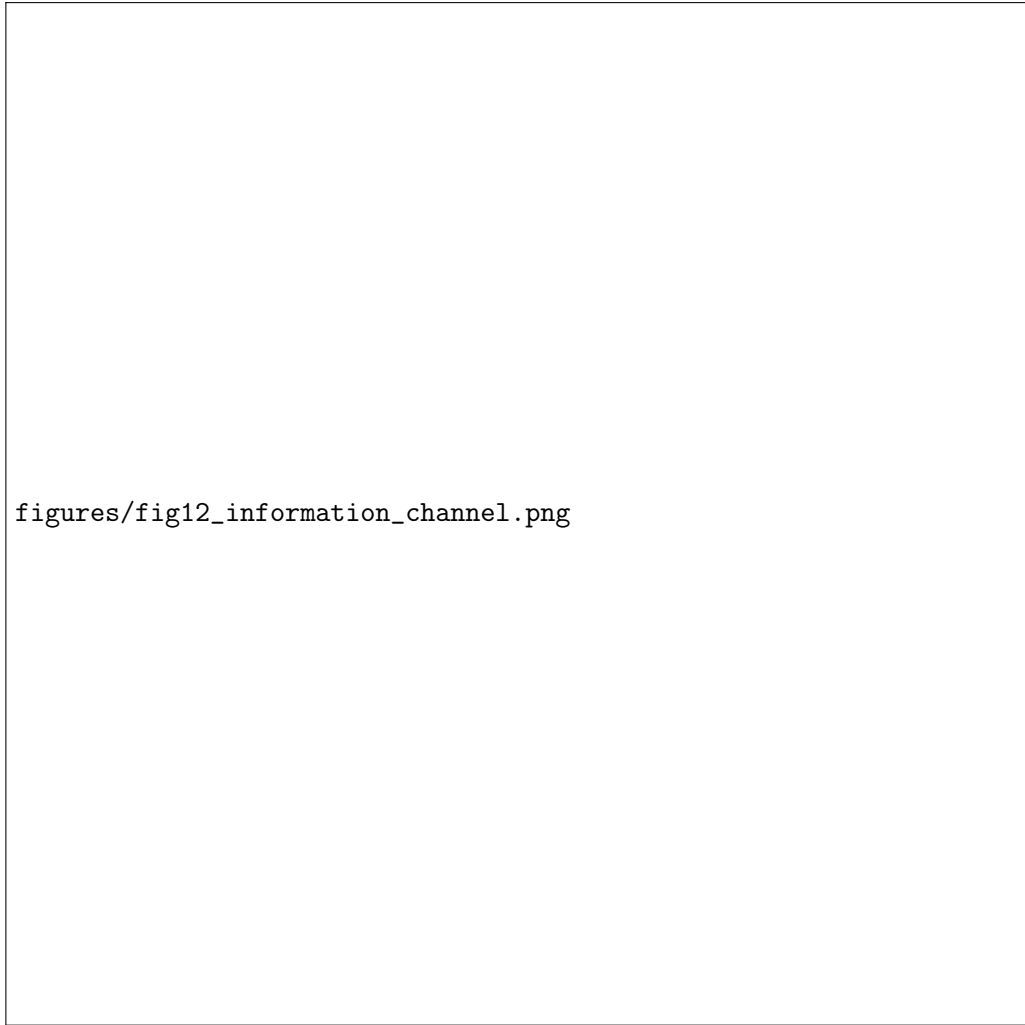


Figure 10: Information channel alignment. (A) CM features preserve the return-domain information encoded in causal edges. (B) GNN embeddings project this information into a latent space, creating a domain mismatch with the return prediction target.

6.2 Empirical Evidence

The empirical evidence strongly supports the mismatch hypothesis:

1. **Feature importance asymmetry.** In the combined CM+GNN model, GNN features capture 37.6% of importance but *degrade* overall IC from 0.203 (CM only) to 0.164. High importance does not imply high *useful* importance: the GBT is drawn to GNN features' variance but this variance is not aligned with the return target.
2. **CM features are individually predictive.** The top CM feature (`cm_relative`, importance = 4.7%) outranks all 34 baseline features. In contrast, no single GNN feature reaches the top 20 in any configuration.
3. **GNN features absorb capacity without adding signal.** Adding 32 GNN features increases model capacity (66 vs. 34 features) but reduces IC from 0.177 to 0.137. This is consistent with the GNN features acting as structured noise that the GBT overfits to.
4. **Causal edges are the best GNN input.** Among the six GNN edge-type configurations, causal edges yield the highest IC (0.173), confirming that return-domain graphs carry genuine signal—but the GNN transformation fails to preserve it.

6.3 When GNNs Should Work

Our analysis does not imply that GNNs are universally inferior for financial prediction. Rather, the information channel alignment framework predicts that GNNs should outperform hand-crafted features when:

1. **The graph encodes non-return information.** Analyst coverage networks, supply chain graphs, patent citation networks, and board interlock graphs encode relationships that are not directly in the return domain. Here, a GNN’s ability to learn latent structural features may genuinely add value because there is no “return-domain shortcut” available.
2. **The relationship between structure and returns is complex.** If the predictive signal lies in multi-hop patterns, community interactions, or nonlinear graph motifs, a GNN may capture patterns that simple aggregation misses.
3. **The feature engineering space is intractable.** For very large or heterogeneous graphs, manually designing domain-aligned features may be impractical, making GNN automation valuable despite some information loss.

The key insight is that *domain alignment should be the default strategy*, and GNNs should be employed when domain-aligned features are unavailable or insufficient.

7 Discussion and Limitations

Limitations of the Causal Graph. Our Granger causality and transfer entropy tests assume stationarity within the 5-year lookback window. Structural breaks, regime changes, and the non-stationarity of financial returns may cause the estimated causal graph to be an imperfect representation of true information flow. The Bonferroni correction for multiple testing is conservative and may miss genuine but weak causal links.

Sample Period. The test period (2025Q1–2025Q4) spans a single year and includes specific market conditions that may not generalize. A longer out-of-sample period with walk-forward validation would strengthen the evidence.

Transaction Costs. We do not account for transaction costs, market impact, or short-selling constraints in our long–short spread calculations. The economic significance of the CM signal in a realistic portfolio context requires further investigation.

GNN Architecture Sensitivity. We employ a single GAT architecture with a specific contrastive training objective. Alternative GNN architectures (GraphSAGE, GIN, heterogeneous GNNs) or different training objectives (return prediction, denoising) might perform differently. However, the consistent underperformance across all edge-type configurations suggests that the issue is fundamental rather than architectural.

Feature Count Imbalance. The CM model (55 features) and GNN model (66 features) differ in feature count. While the GBT should be relatively robust to irrelevant features, the 32 additional GNN features may introduce a regularization burden. We note, however, that the combined model (87 features) also underperforms CM-only (55 features), and the GNN features receive substantial importance weight, arguing against the explanation that they are simply ignored.

Overfitting Considerations. The training IC of 0.880 versus test IC of 0.203 suggests meaningful overfitting in the GBT, which is typical for tree-based models on financial data. The key comparison, however, is *relative* test performance across configurations, which is less susceptible to overfitting artifacts since all models share the same train/test split.

8 Conclusion

We have introduced causal momentum features—a family of cross-sectional signals derived from Granger causality and transfer entropy graphs—and demonstrated their superiority over GNN-based graph embeddings for stock return prediction. The CM model achieves an IC of 0.203 versus 0.177 for the baseline and 0.137 for the GNN model, with the single most important feature (`cm_relative`) being a CM feature.

We explain this result through the *information channel alignment* hypothesis: when a graph encodes return-domain causality, predictive features should remain in the return domain. GNNs, by projecting graph information into a latent embedding space, create a domain mismatch that destroys predictive signal.

This finding has practical implications for quantitative researchers: before deploying a GNN on a financial graph, one should ask whether simple domain-aligned feature engineering can capture the same information more directly. Our results suggest that for return-causal graphs, the answer is yes—and that the simpler approach not only matches but substantially outperforms the neural network alternative.

More broadly, our work contributes to a growing recognition that domain knowledge and thoughtful feature engineering remain competitive with—and often superior to—end-to-end deep learning approaches in structured prediction settings where the relationship between inputs and targets is well-understood.

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